Study Guide
A Linux VM Installation Guide

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Abstract

This document is the study guide for the first week. In week one, we will use need to get a Debian-based Linux such as Ubuntu, Kali linux installed on our computer. Fedora-core linux is also accepted.

Linux operating system is the base operating system we will use to conduct this course in this whole semester. If you encounter any issues, please stay in the classroom and I will help you to resolve your problem.

Note: The screenshots and written part are subject to changes due to frequent software updates and dynamics in IT field.

1. Install Debian-based, Ubuntu 22 LTS, Ubuntu 24 LTS or Kali Linux. This is a required step to begin learning the linux commands and systems. I also released our first homework early and as you can proceed with those questions as references for what to study.

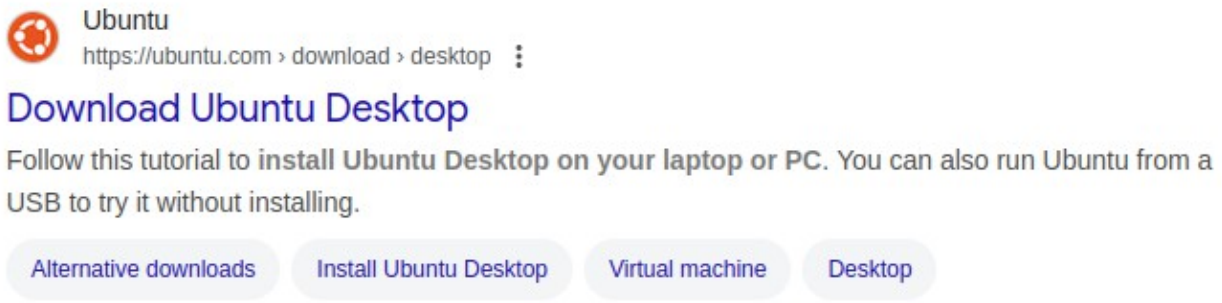
In this part, I am going to give you a little guideline to help you get linux installed and started.

Note: If you are a linux user and already had linux as your main operating system or have linux installed in a VM, then you can ignore this section.

For those of you who are beginners and who did not have linux before, you would need to follow the guide below,

Step 1 Download Linux ISO file

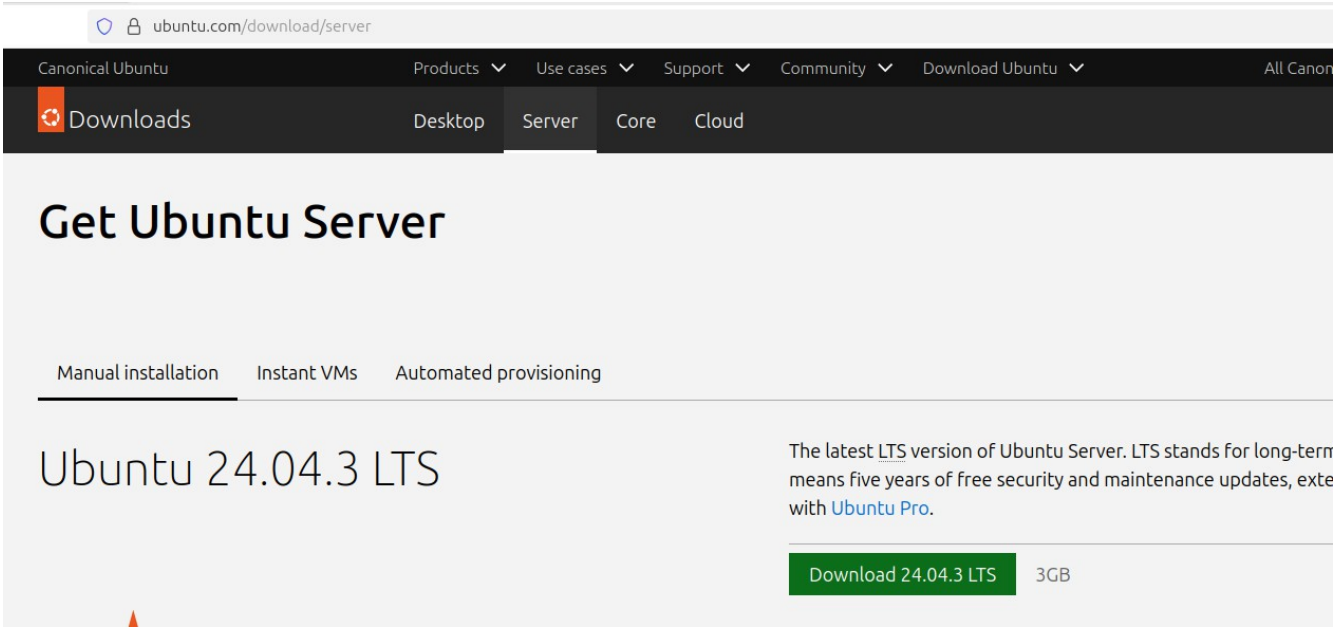
Normally, there are desktop versions and server versions. The following is a desktop version, for instance.



And the following one is for server. I would recommend we click the following link to download a server version,

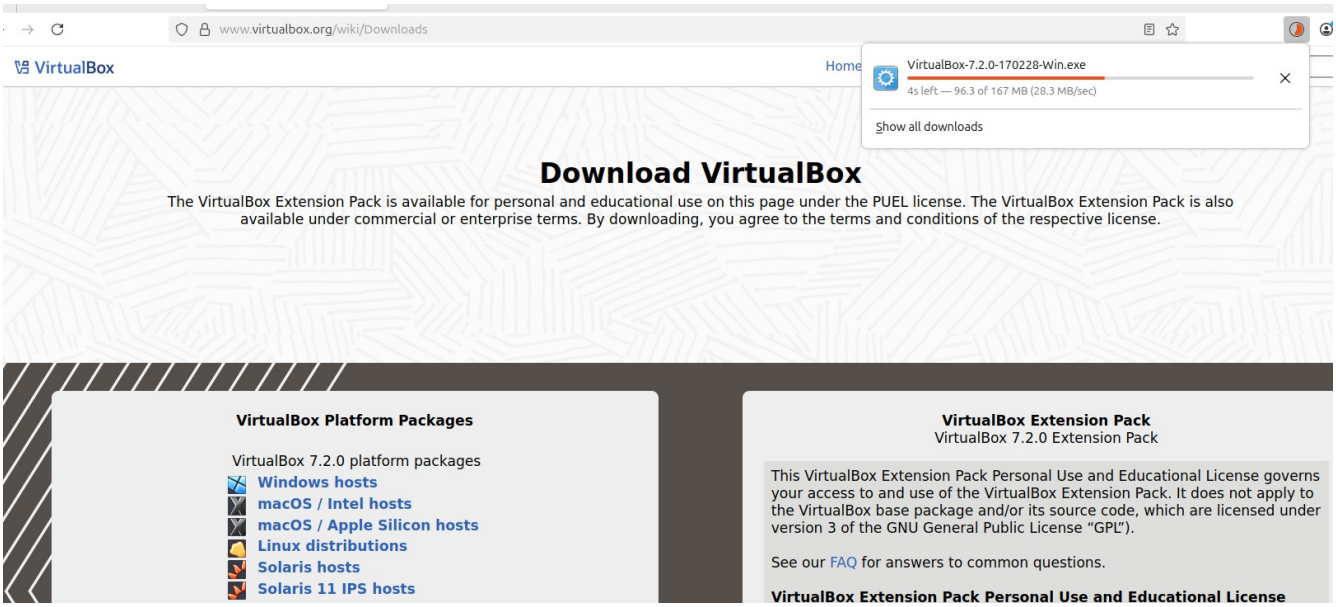


In the following page, click the green button and you will download the ISO file.



Step 2 Download and Install VM emulator on your machine

For Windows users, please use the Oracle virtualbox,



Follow the default options to complete installation of the virtualbox.

Here I listed a couple youtube videos on how to install virtualbox on windows 11.

https://www.youtube.com/watch?v=JWdxMZ2ldYA

https://www.youtube.com/watch?v=homRENM8KVY

Step 3 Install Ubuntu with VirtualBox

Please follow the video blow to complete the Ubuntu installation with Virtualbox using the Ubuntu ISO we downloaded earlier.

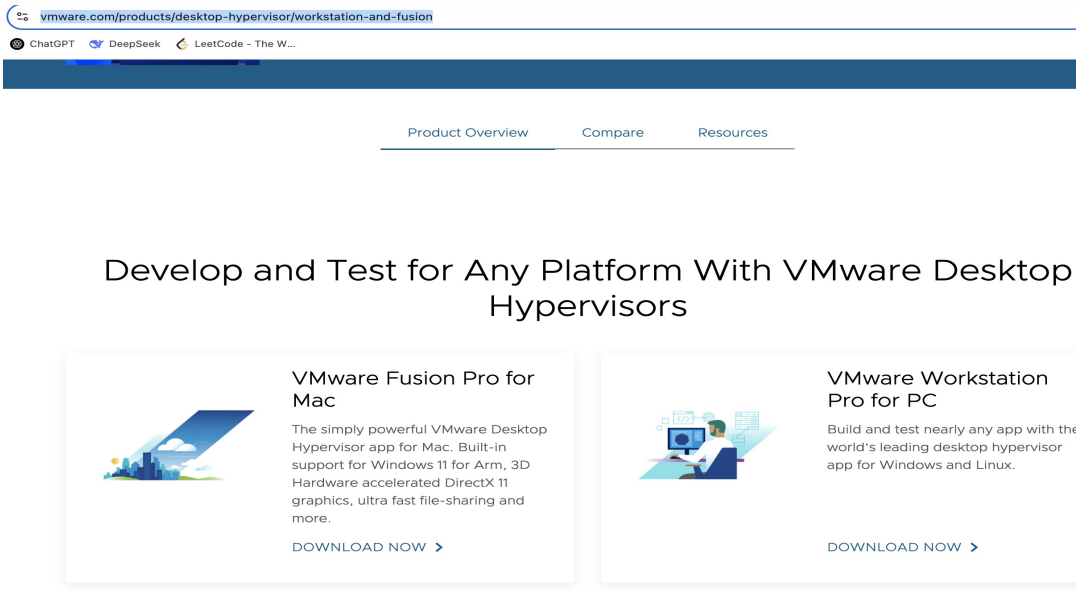
https://www.youtube.com/watch?v=Hva8lsV2nTk&pp=ygUjaW5zdGFsbCB1YnVudHUgaW4gb3JhY2xlIHZpcnR1YWxib3g%3D

For Mac Users

Please go to the following url to download and install VM fusion. I will provide a little more detailed instructions on the installation of both fusion and linux as Mac and recent M chip (M1-M3) Mac are using non-intel CPUs and the assembly/machine code had a lot of software compatibility issues.

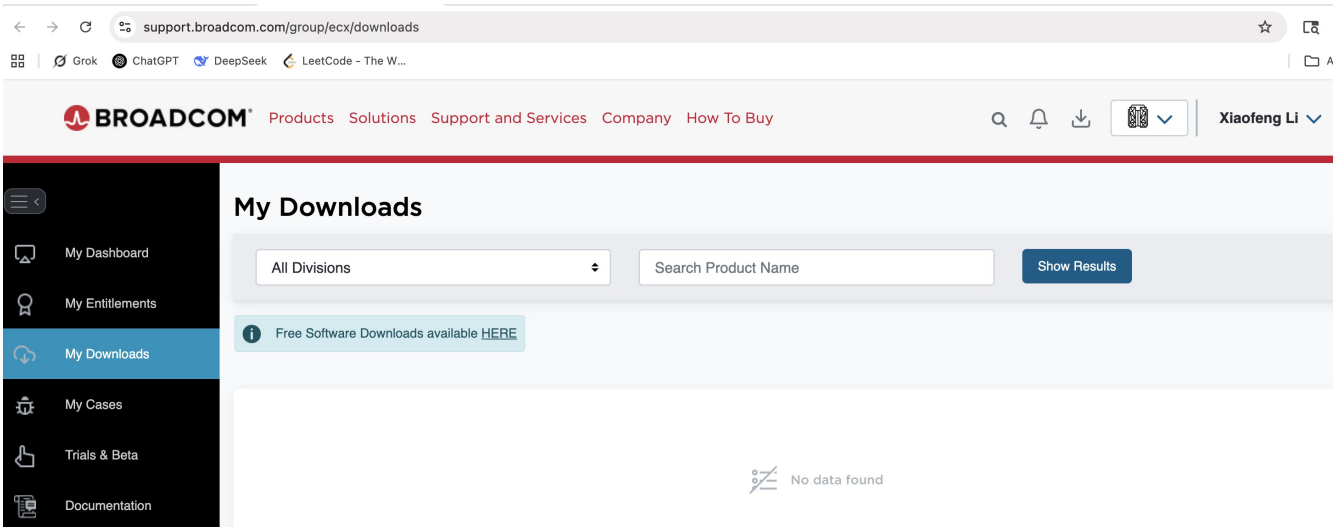
All the instructions in the following section was done in an Intel-Mac Pro, not M chip. From the fusion I downloaded, I think it would support M chip as well but I did not test it.

https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion

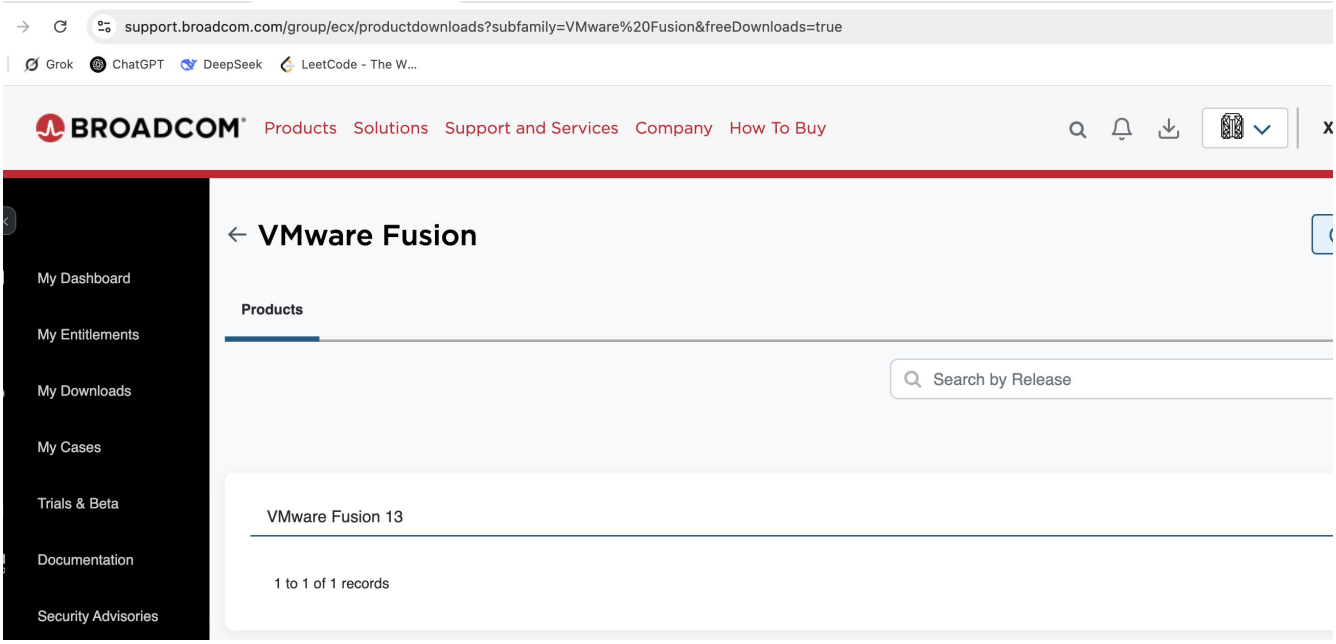


Click the button “Download Now” on the left like above. You would need to register a free account before proceed to download.

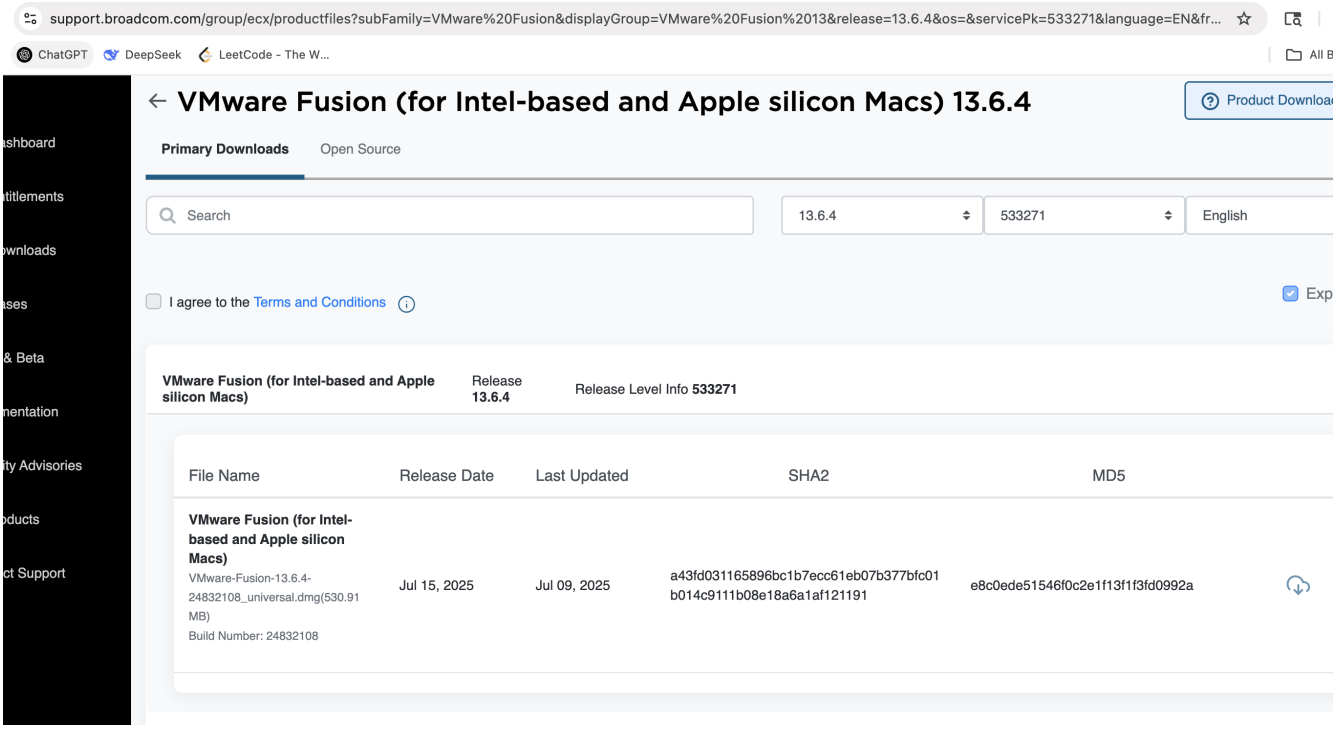
Please write down and remember your password, and you will go to login and see the following dashboard. Note, and click on the blue line, there is a link “HERE”,



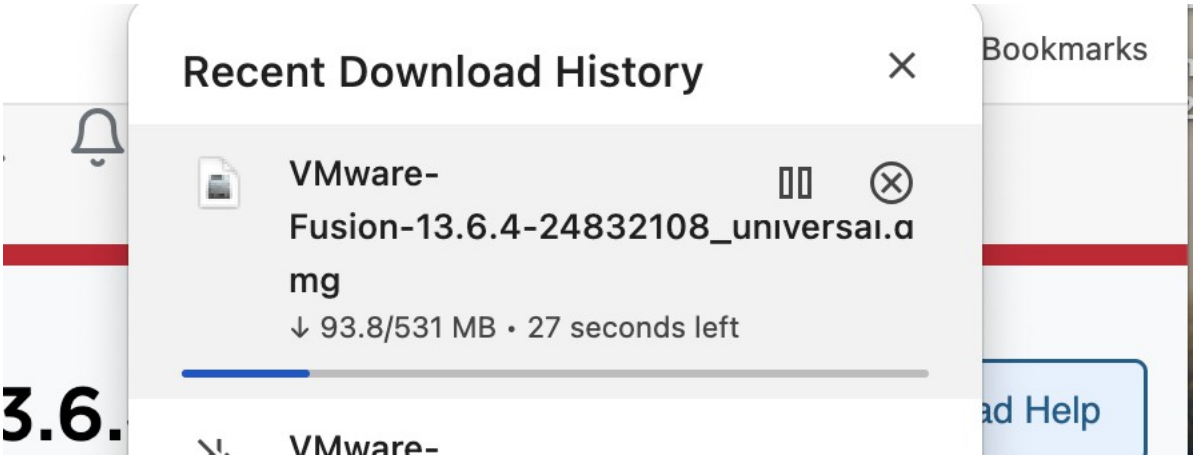
Then you will see a list of free software,



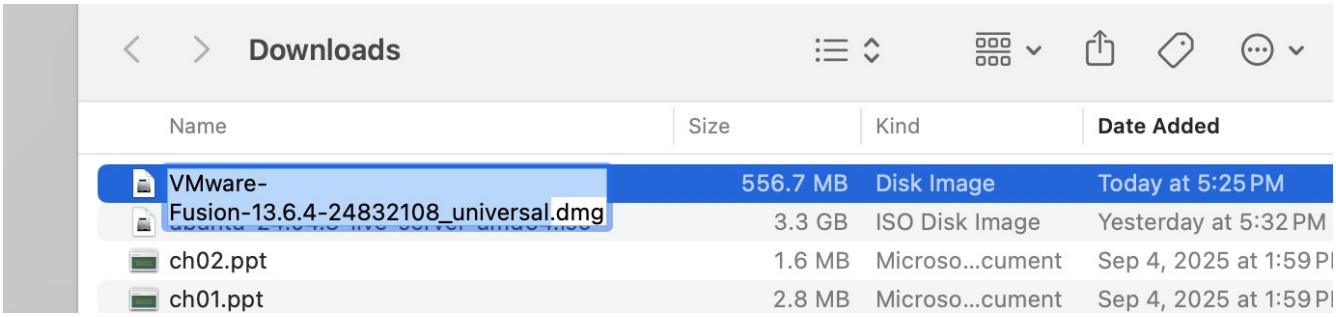
Next,



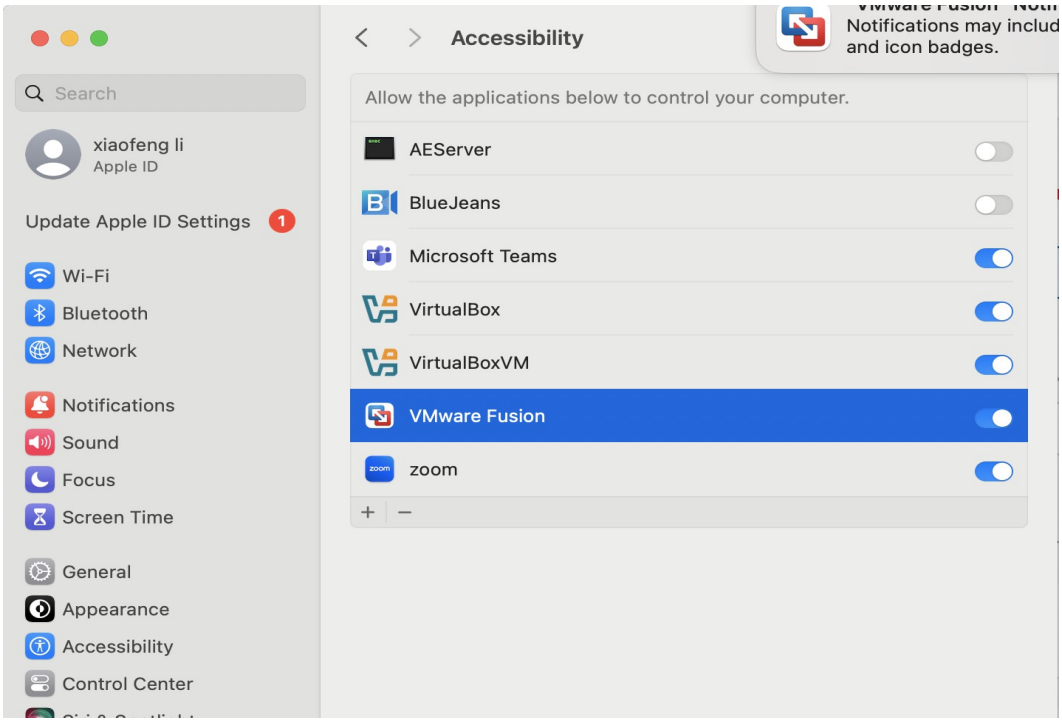
You need to click on the “Terms and conditions”, and then check the box and then follow downloading,



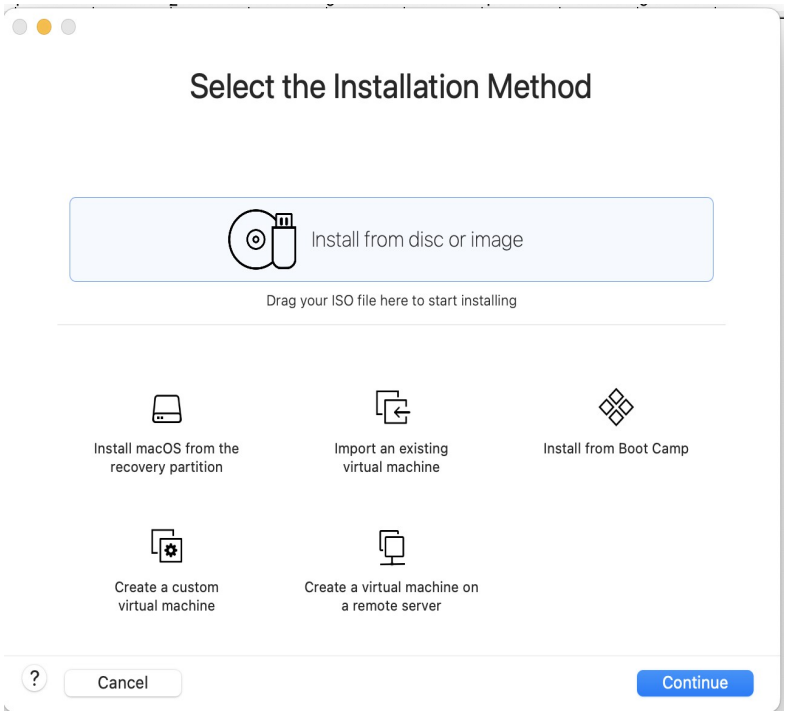
Install it in Mac by double clicking the dmg you just downloaded,



Follow the Mac installation and turned on in security configuration in Mac,

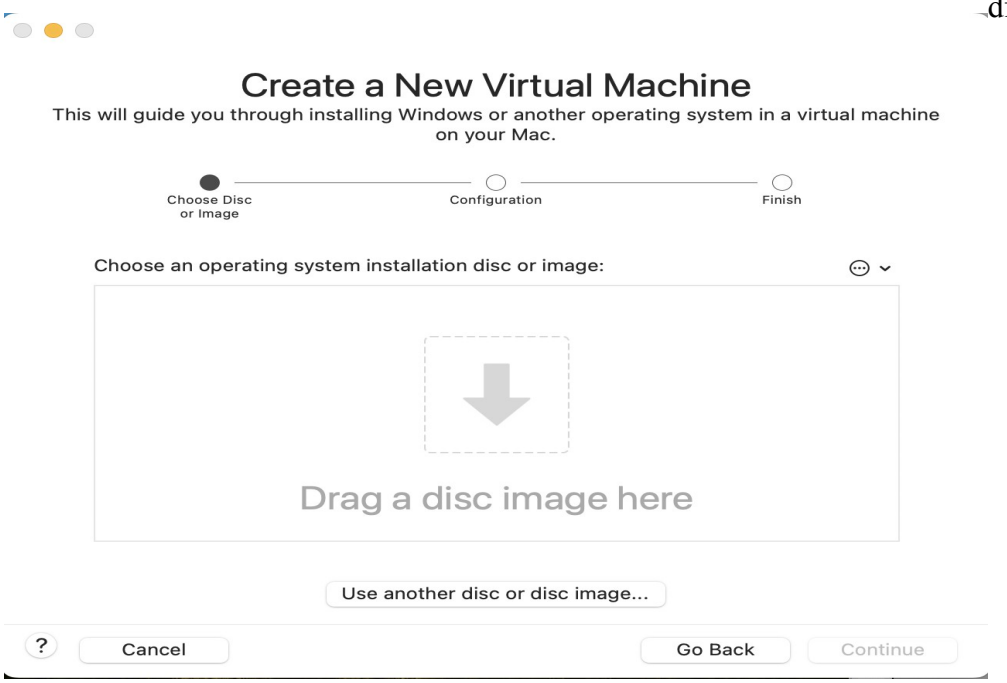


Now the VM Fusion looks like the following,

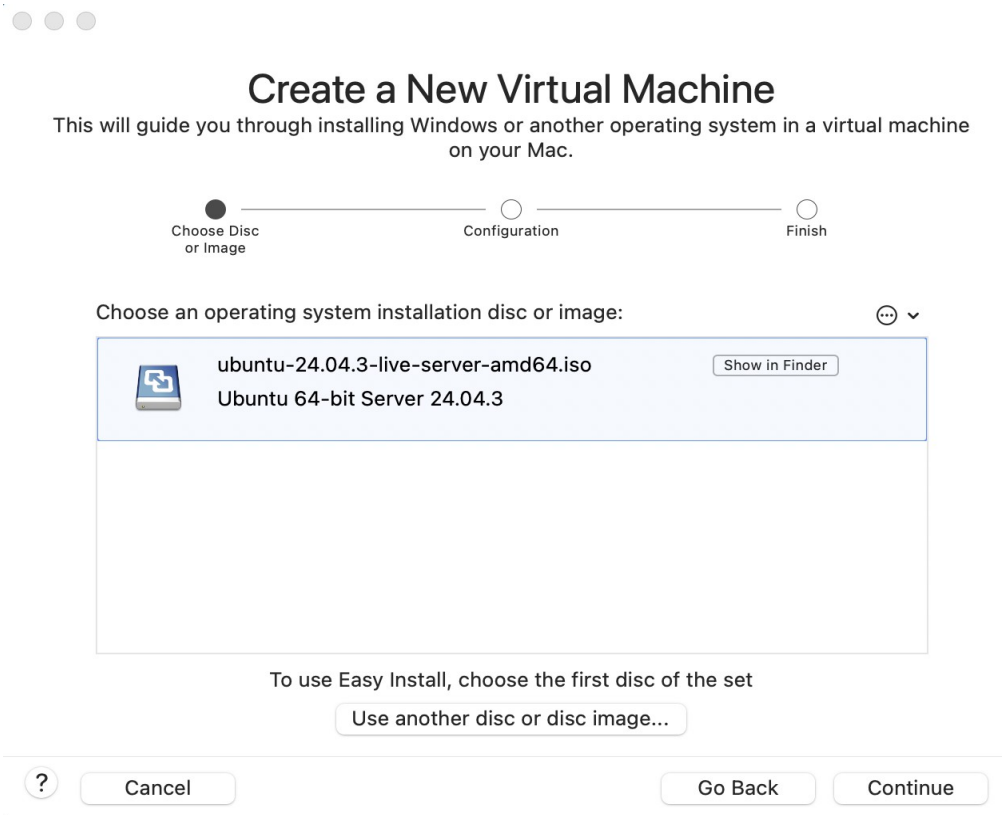


Click the disk icon above, and you will see the next screen in the following image,

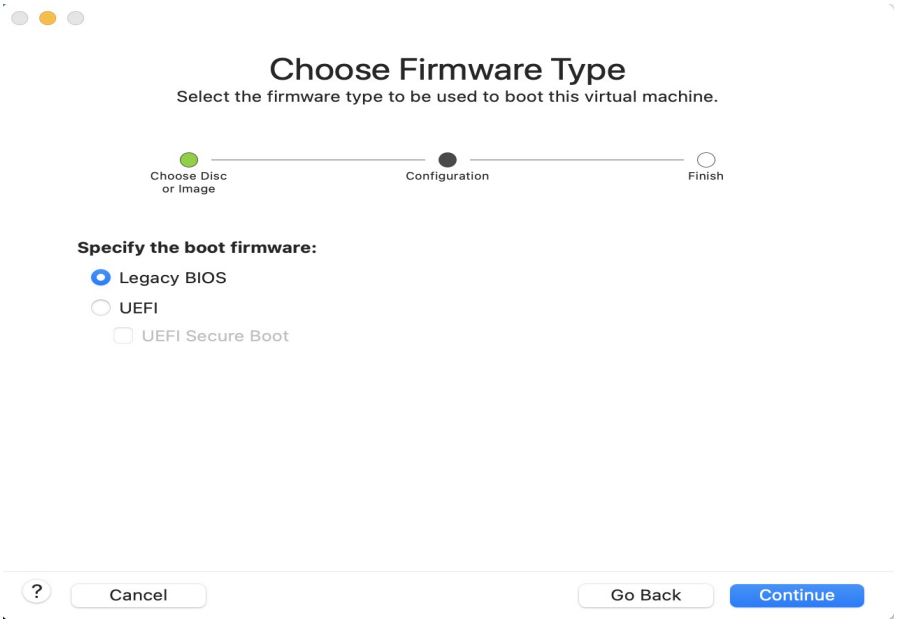
df



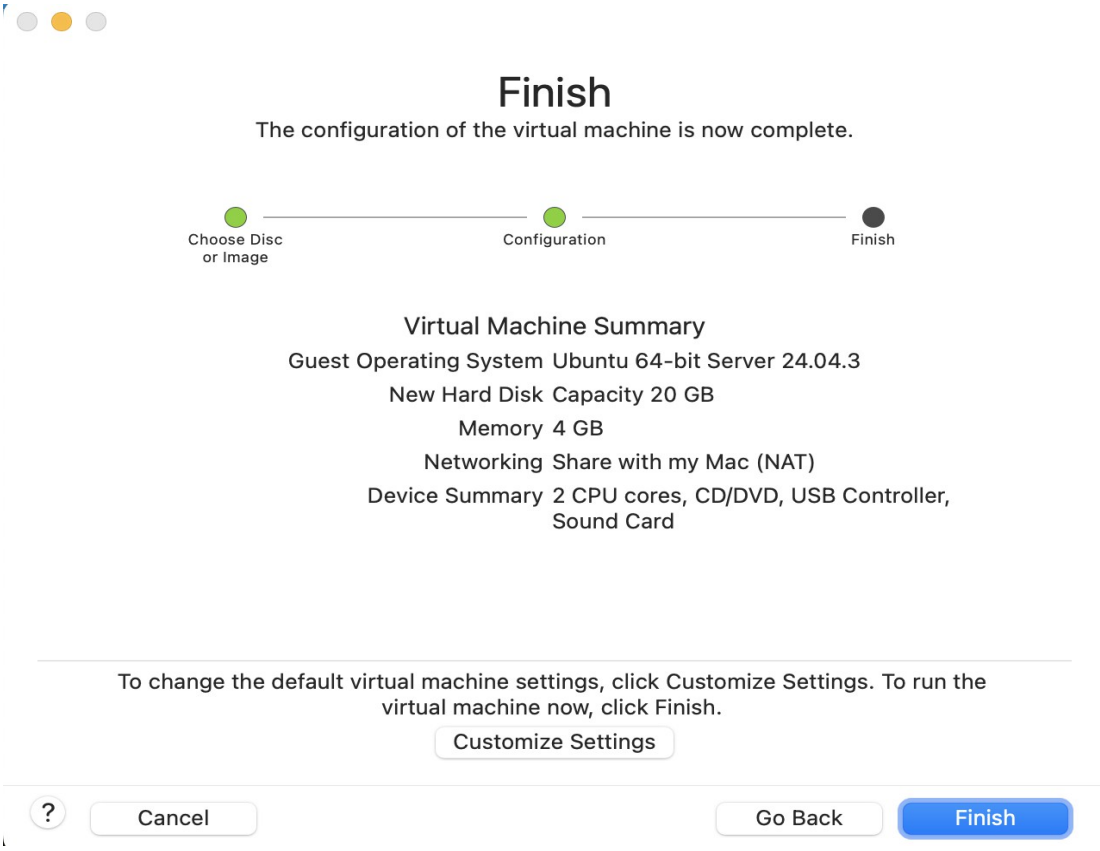
And drag an ISO you downloaded, in my case, I had a ubuntu 24 server ISO,



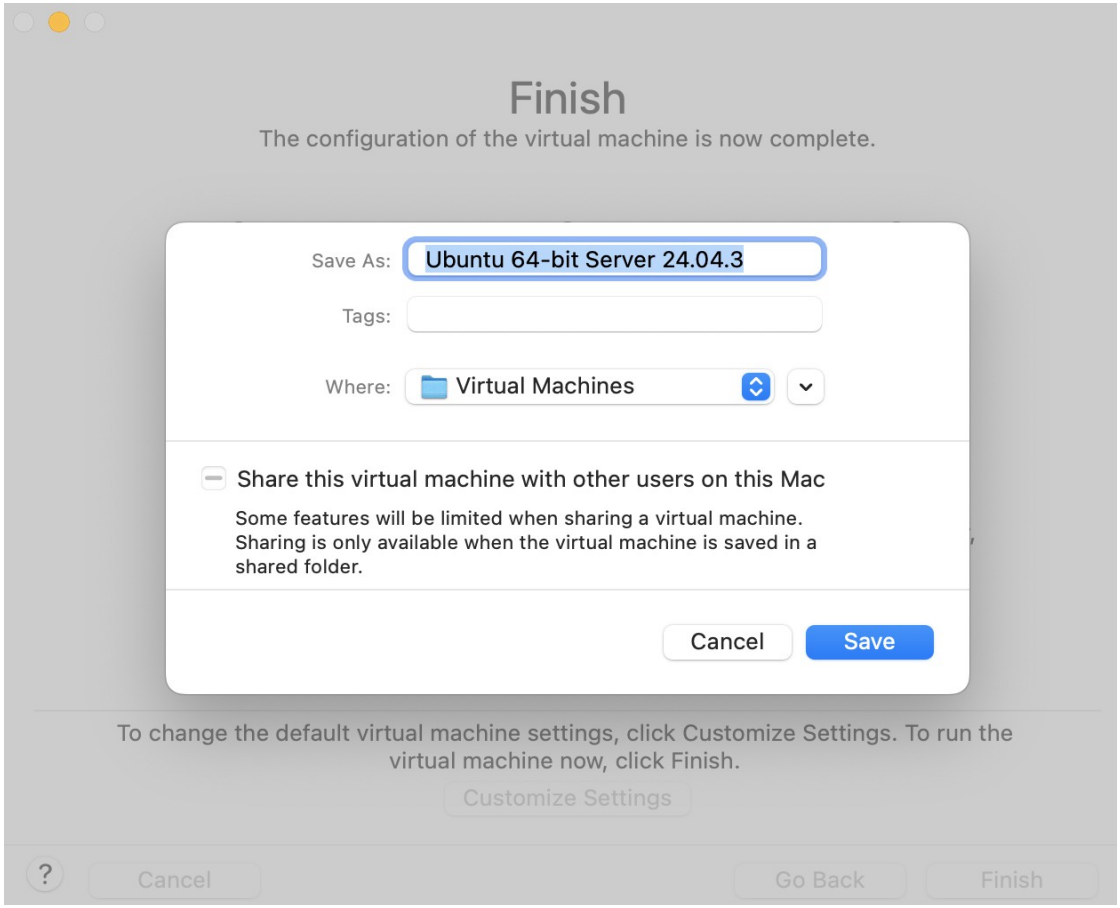
In the following screen, keep the default option selected, and proceed with “continue”,



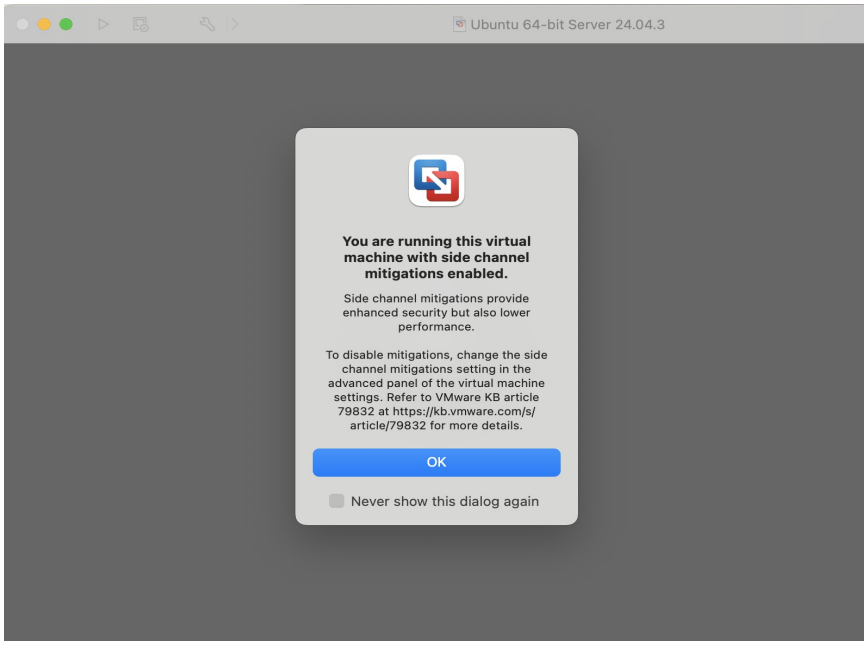
Next you will see the screen,



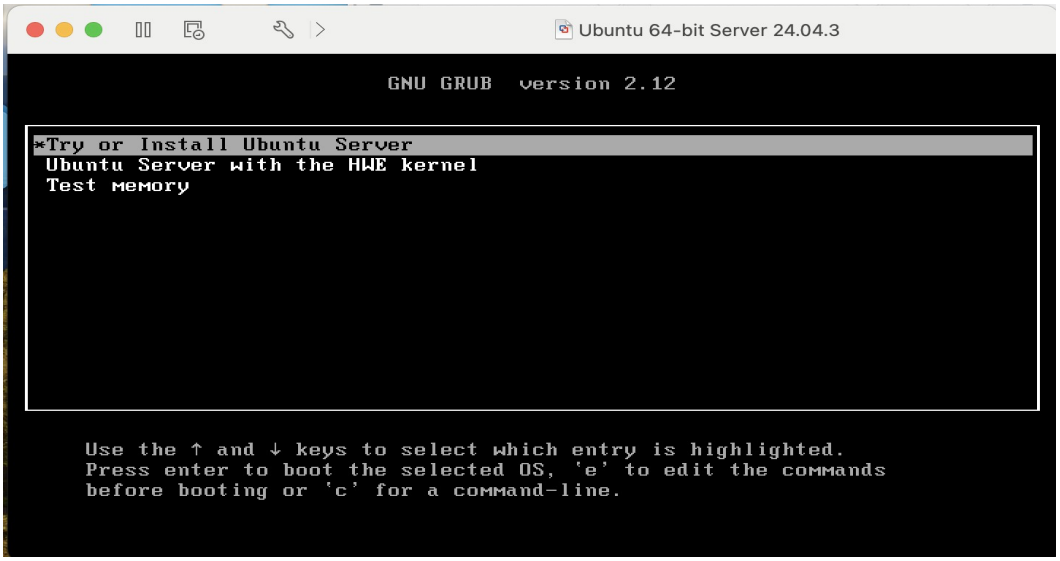
Click “Finish” above, and you will see the image below. This prompts you to save the image to a directory. Note that an VM image file is what we just created, not the Linux installer ISO.



After that, you will see it automatically starts like the following,

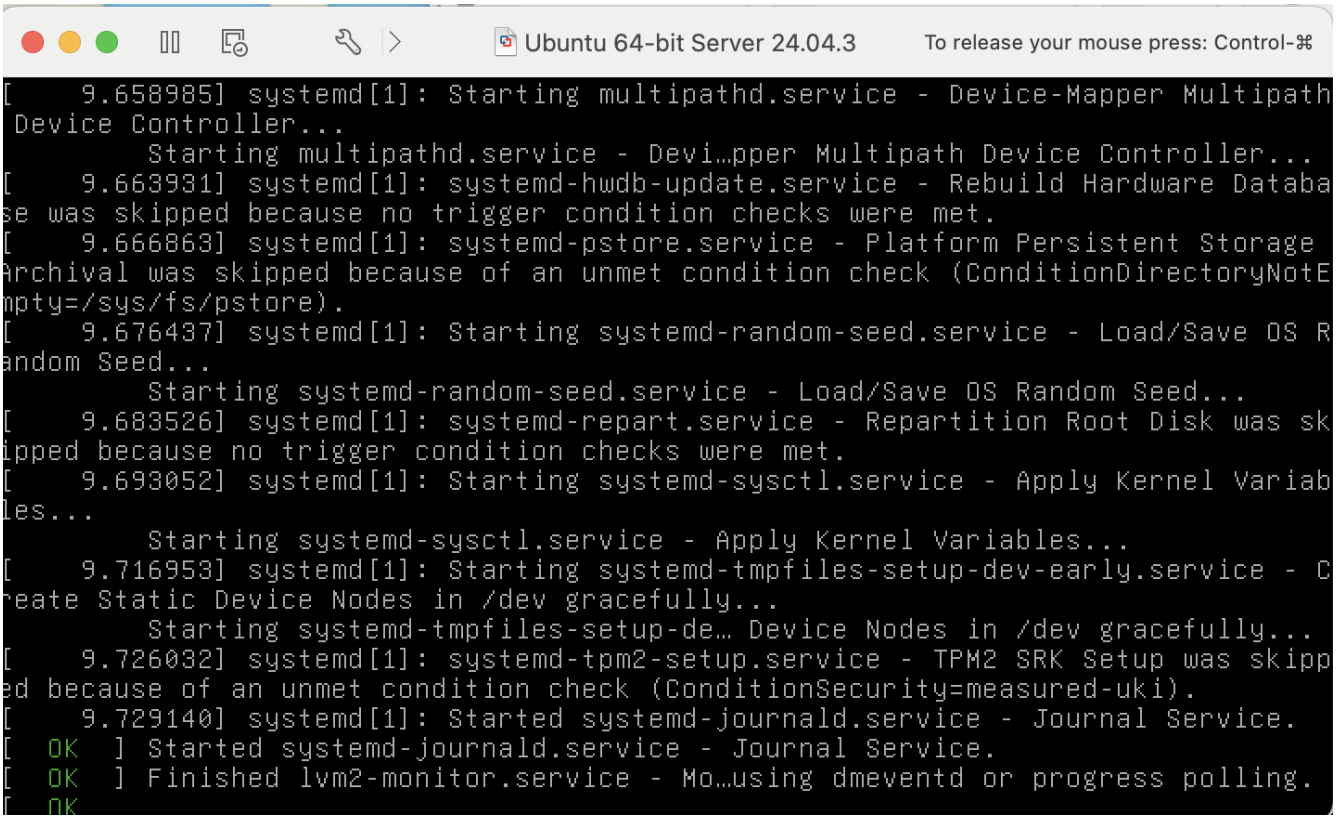


Next, you will see a play button, and it comes to the following screen,

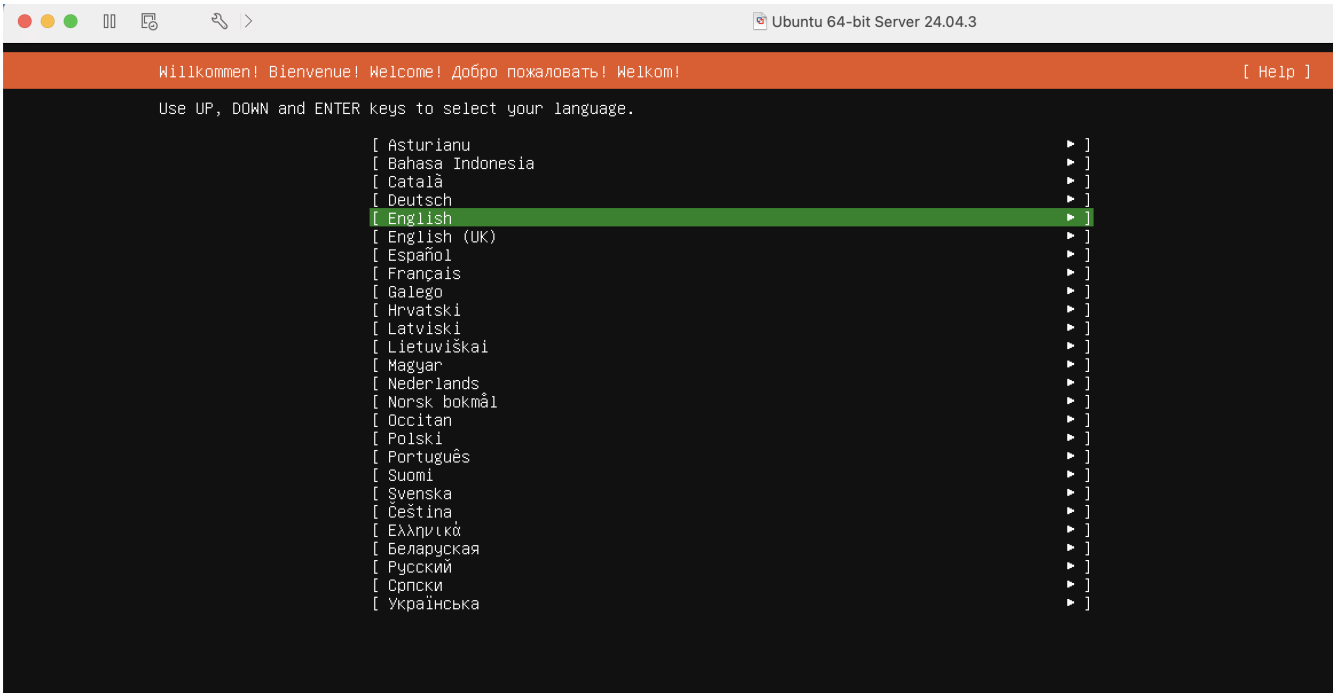


Now up to here, we just finish create the vm image, and we boot/launch the vm image, it is as if we put the Linux installation DVD disk into the DVD ROM, and the machine boots from it. It tells us that, from now on, we are firstly installing the real linux operating system into the VM image file and save the states after all of this will complete. Then by then, your VM image will become a bootable linux server OS.

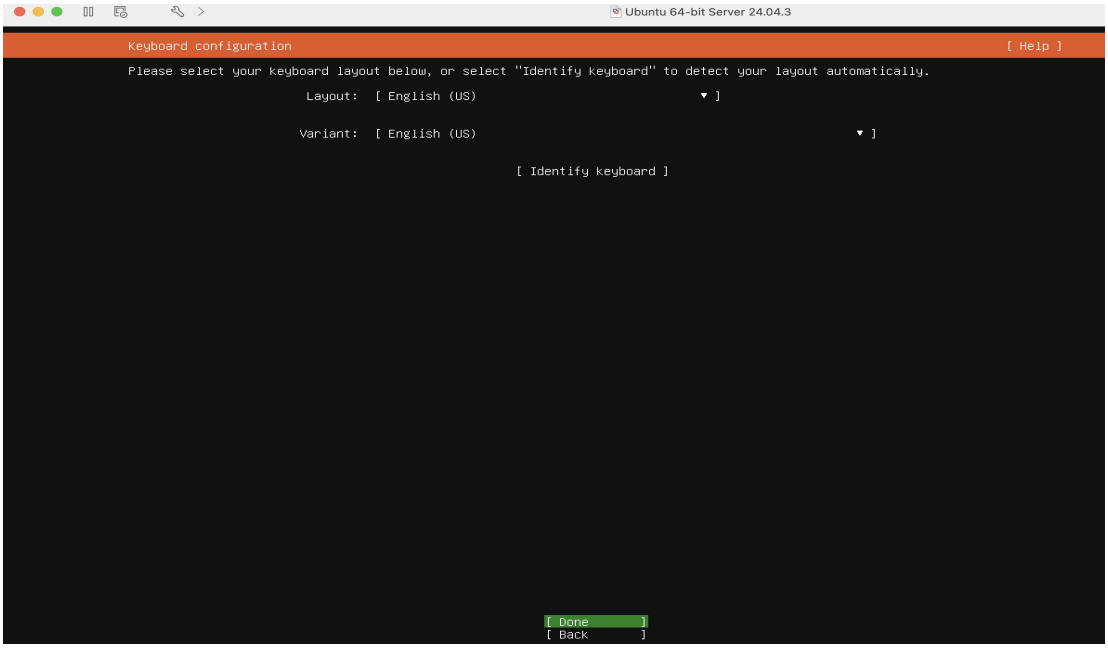
Choose the first option, and hit enter.



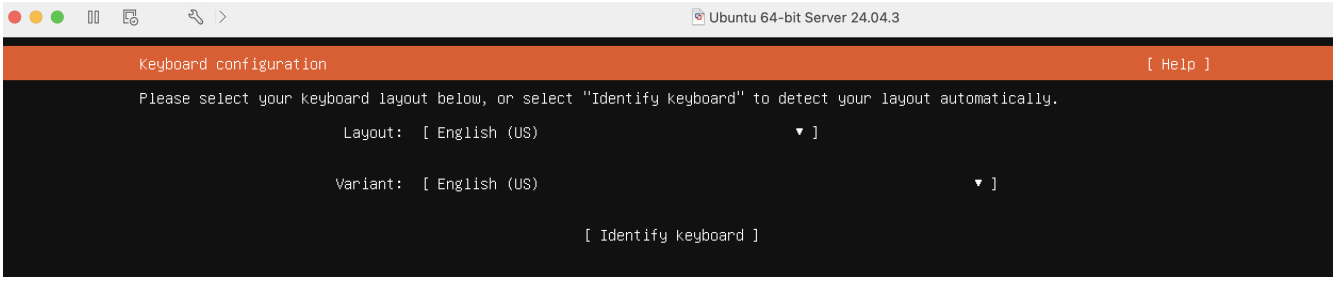
Now, we will follow each screen to finish the installation till the end. (Keep all the default options, and hit done.)



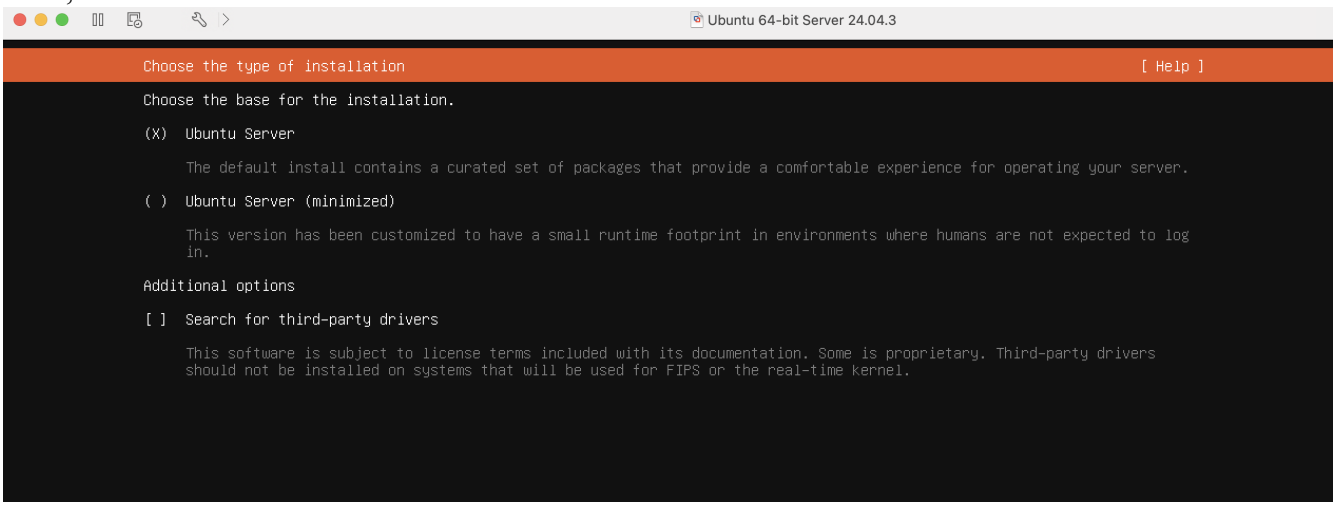
Next screen,



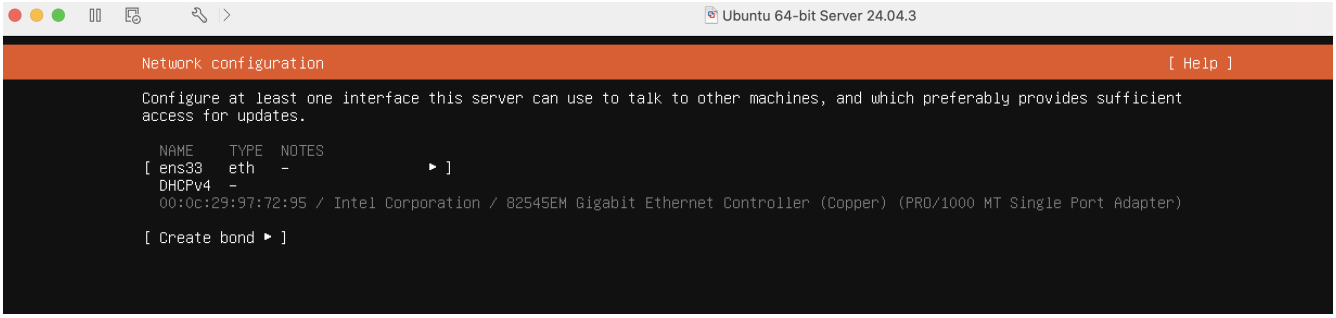
Next,



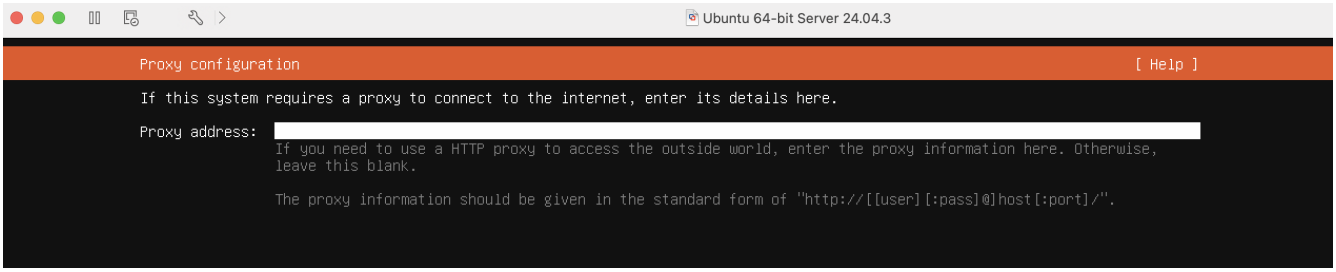
Next,



Next,



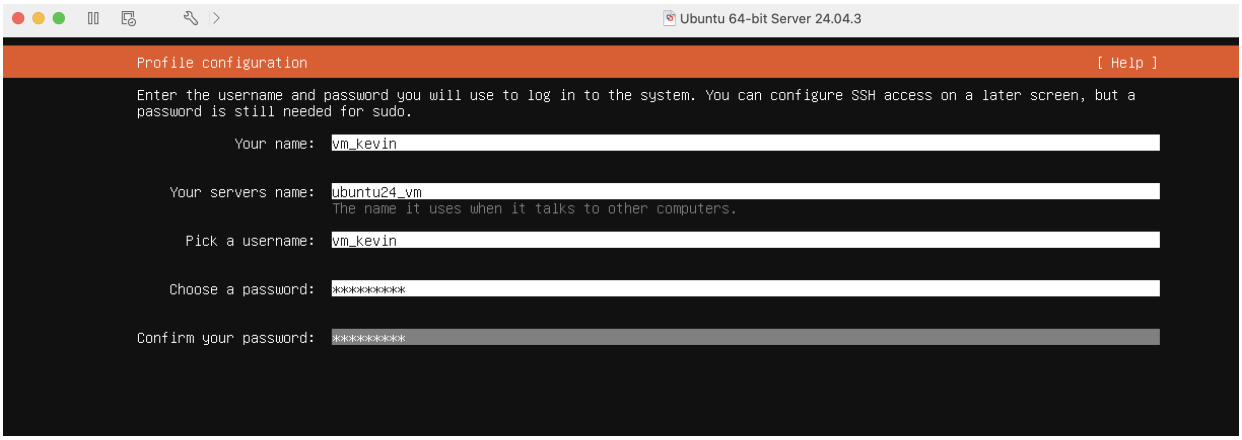
Next,



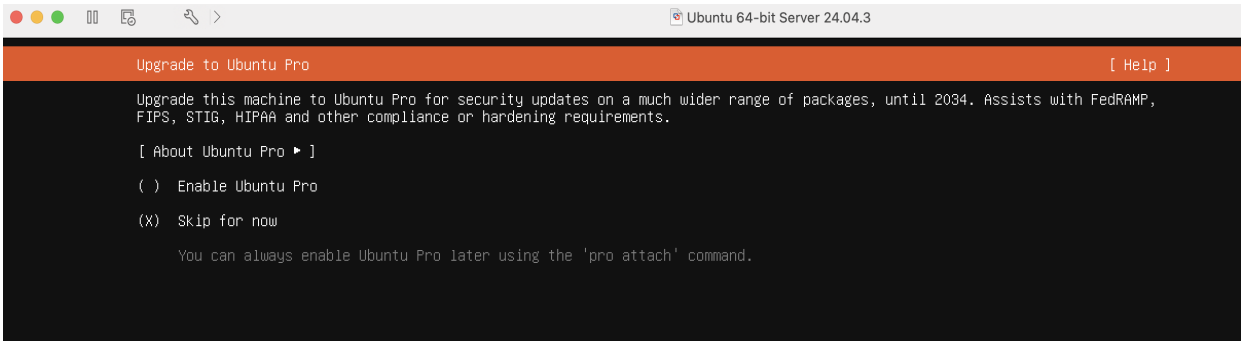
Next,



Next, use your “tab” key on the keyboard to fill them out, please write down your username and password.

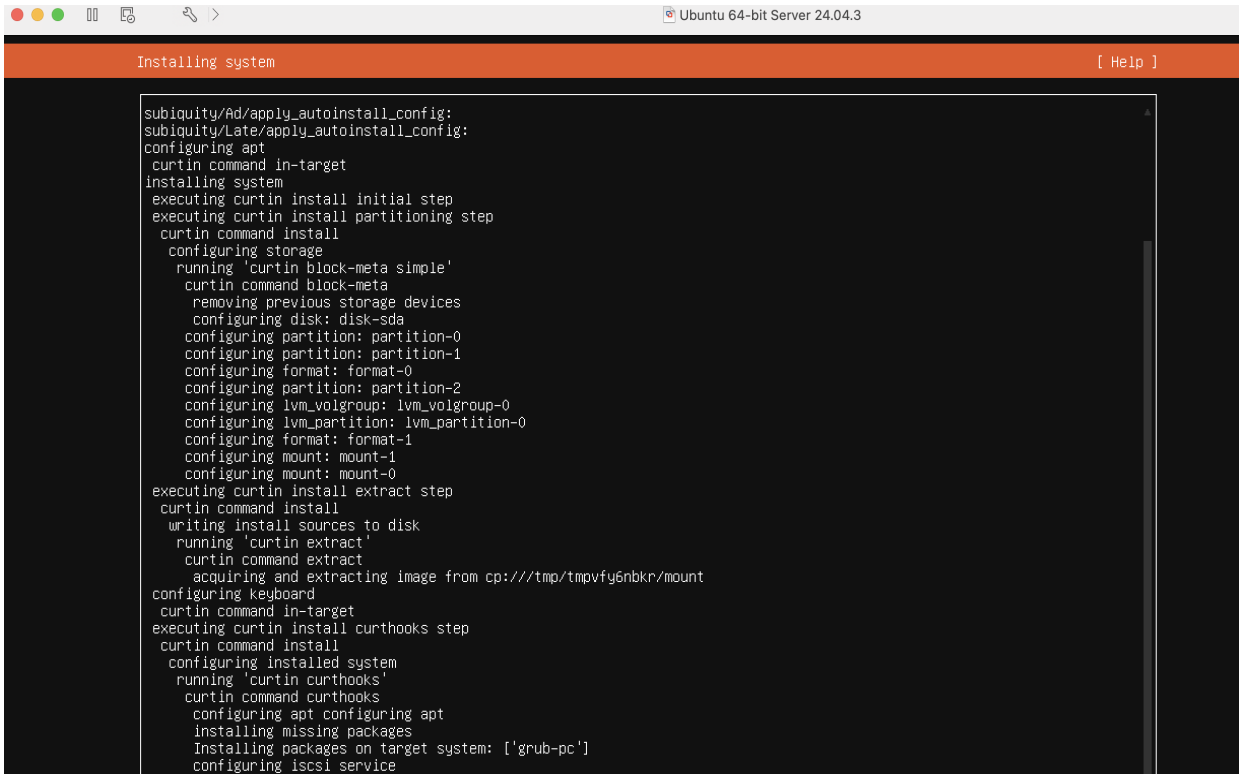


Next,



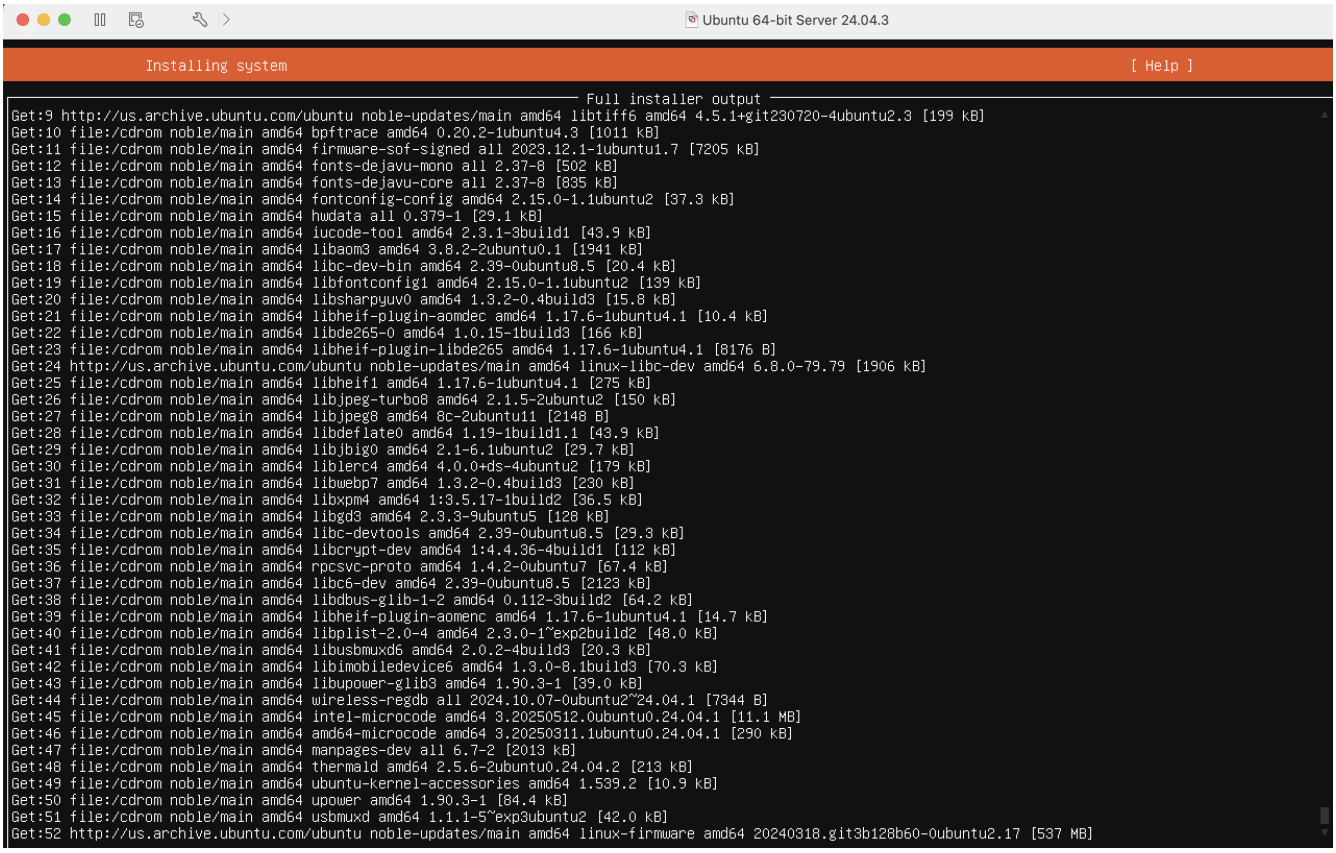
Note: In order to get to the “Done” or “Back” button, use tab key on your keyboard.

Wait for a little bit,

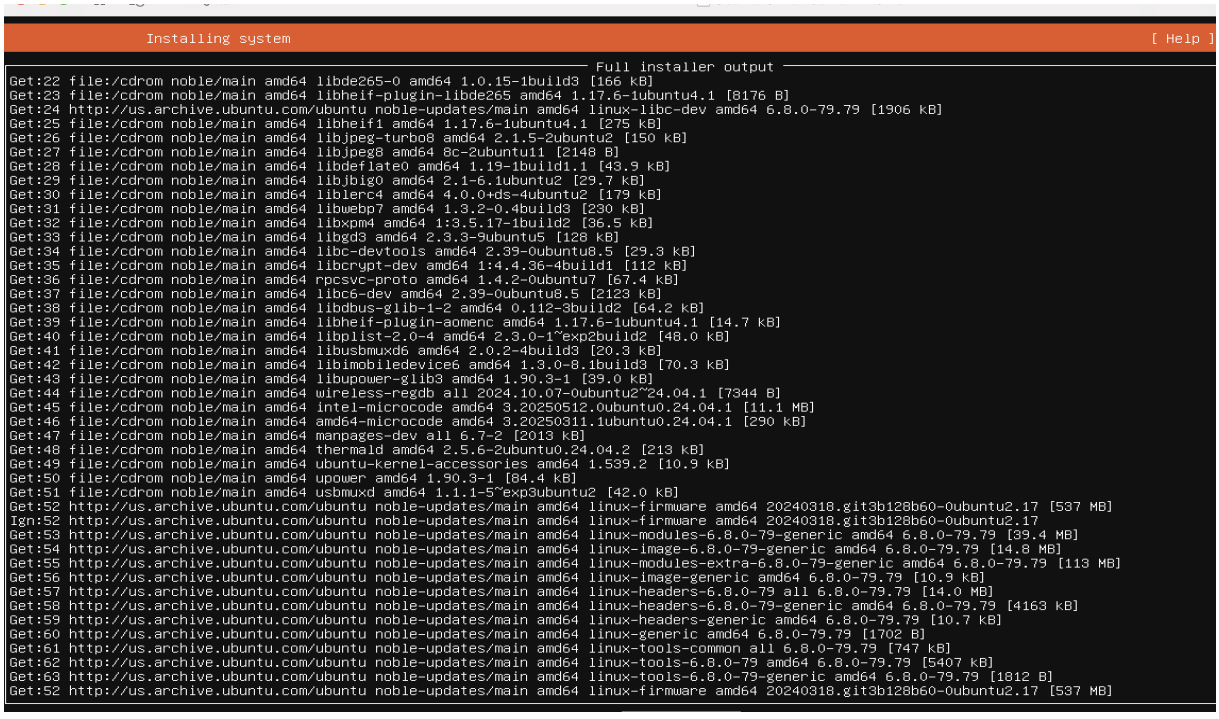


This will take quite a while.....

If you wonder what was going on behind the scene and why it is taking so long, you can hit “View Full Log” in the bottom. You can see the item number there. Some might be downloading so it take some time.

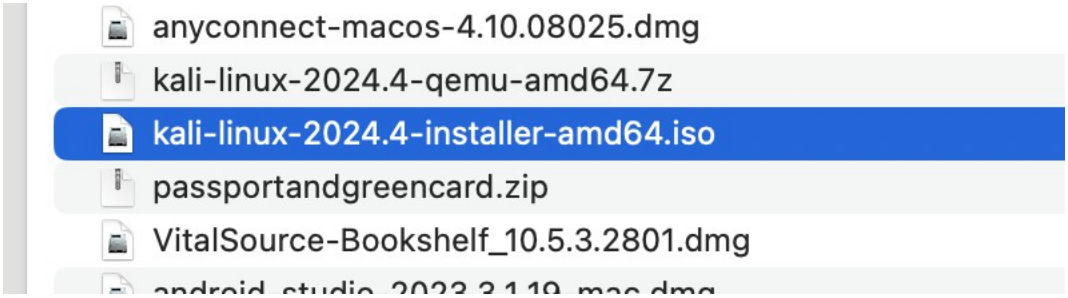


New updates,

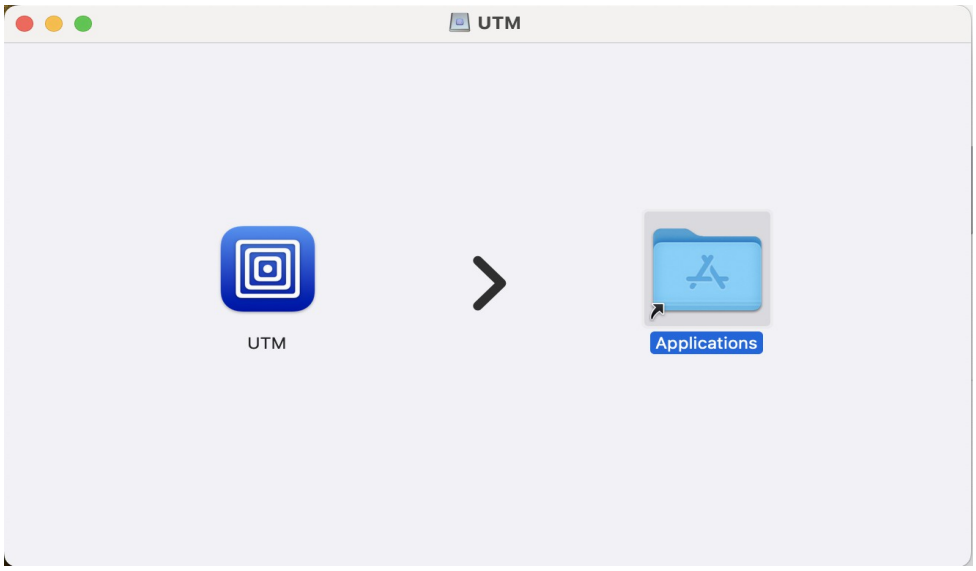


UTM VM

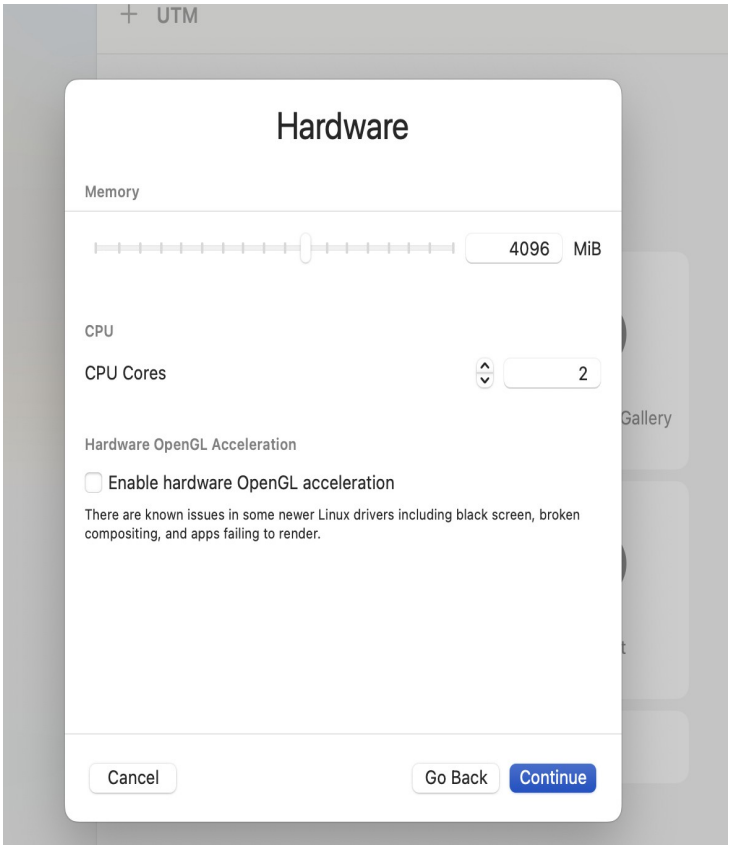
If this crashes or giving you other troubles, we can use another VM.
In this example, I am using Intel Mac and install a kali-linux-2024.4-installer-amd64.iso.

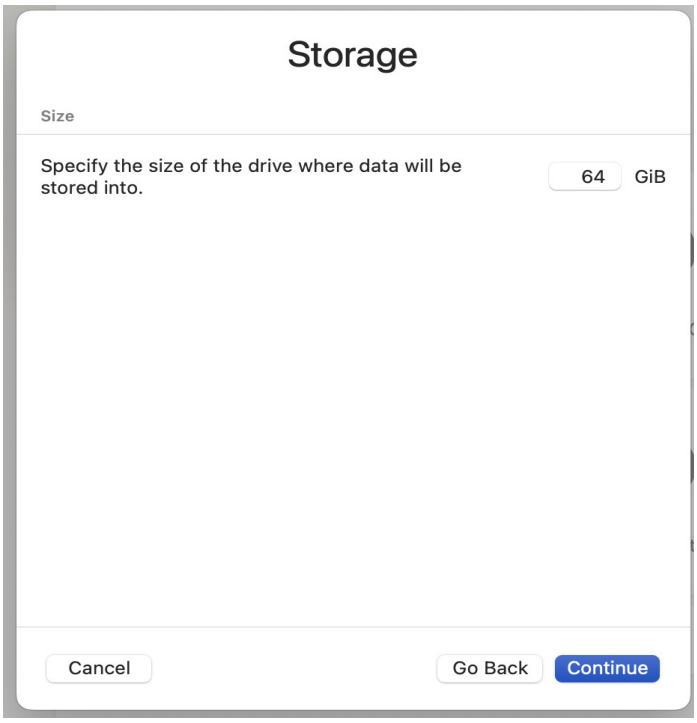


First install the UTM in your Mac,

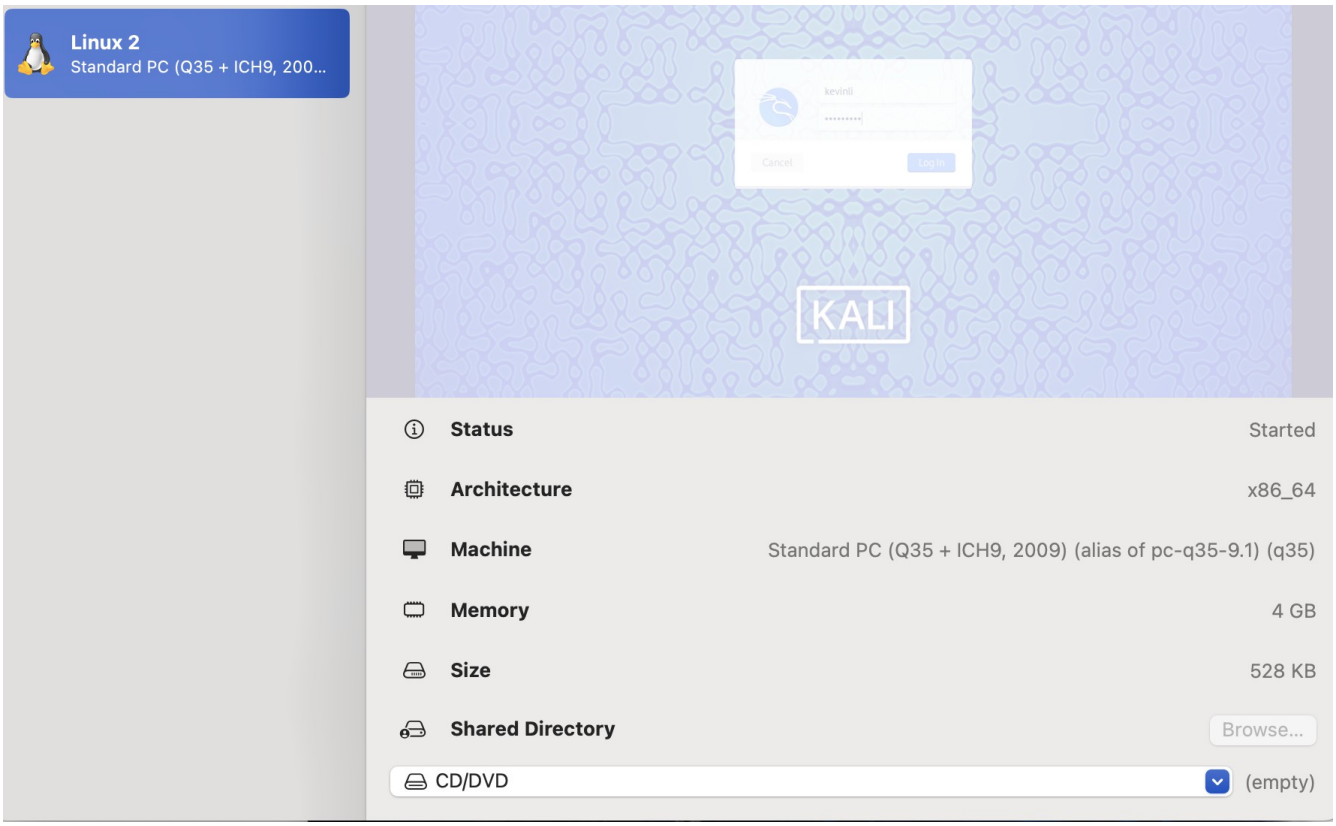


Then, create a new VM, and follow the instruction to finish installing a linux ISO.

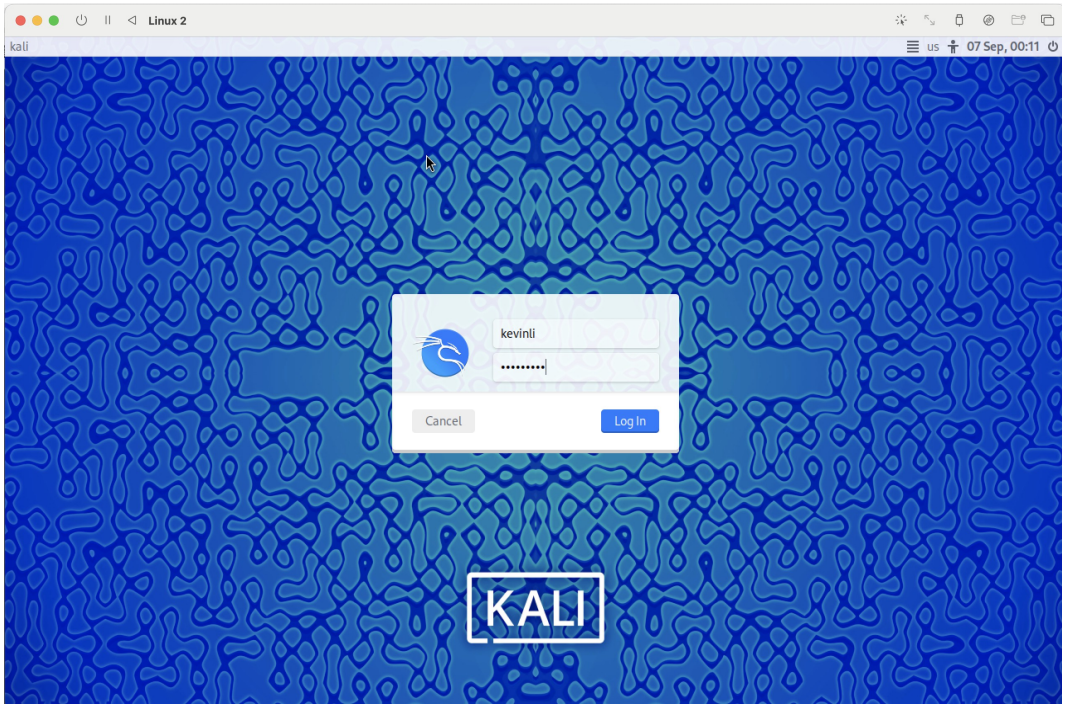




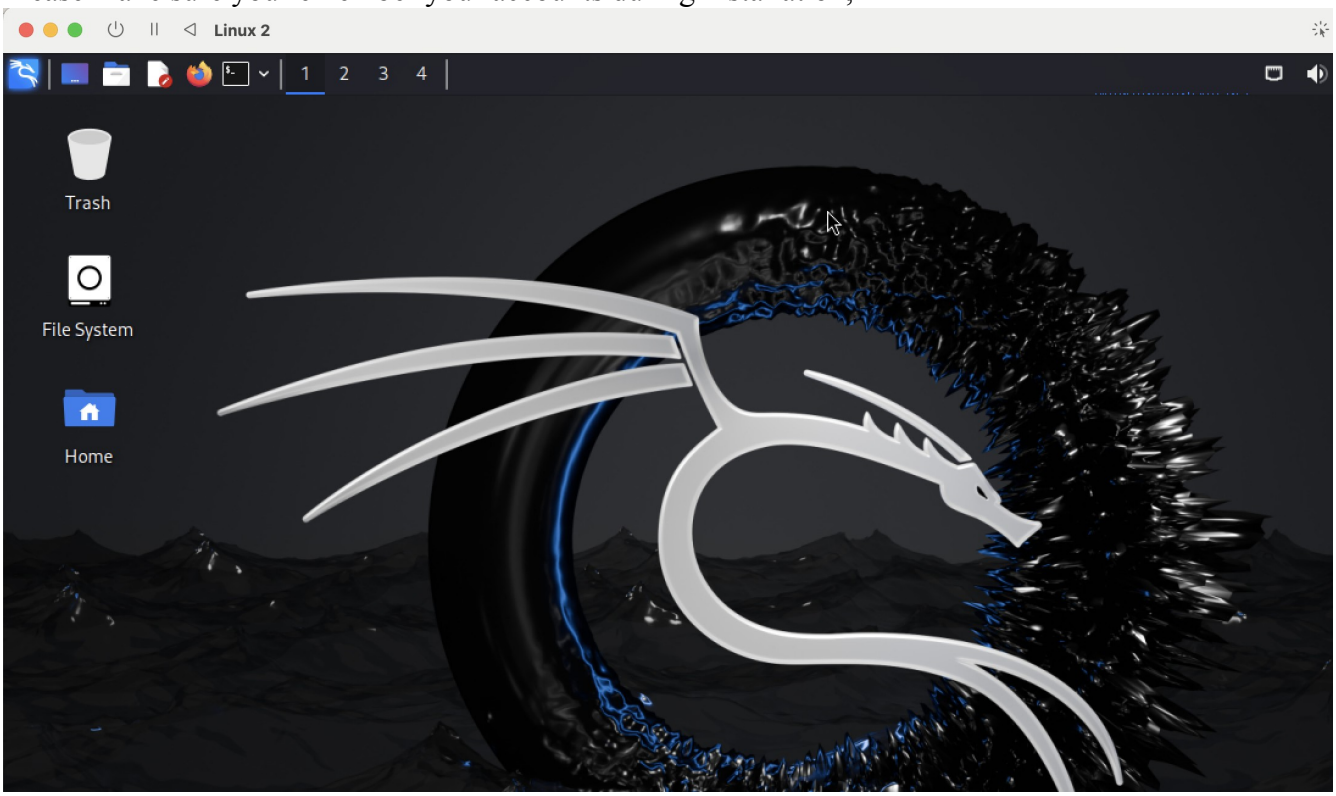
After the installation, you will need to remove the ISO file for the VM image. In the bottom of the CD/DVD, please click “clear”.



Power off the VM image, and reboot.



Please make sure you remember your accounts during installation,



Now it is all good, and we can start a CLI/shell terminal to start learning the linux commands.